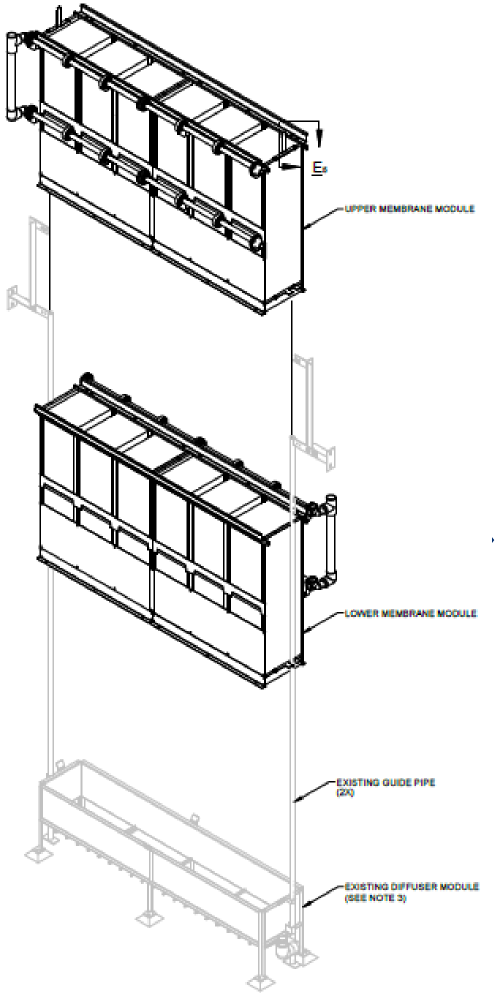
Replacement Procedure

'TORAY'

Disconnect	Disconnect filtrate and air scour hoses
Remove	Remove Competitor K element block. Keep air scour block in place.
Install	Install Toray NHP element block on pre-existing Comp K air scour block.
Reconnect	Reconnect filtrate and air scour hoses. (Modified hoses may be required.)
Resume Operation	Return to normal operations.

Before

Competitor Module



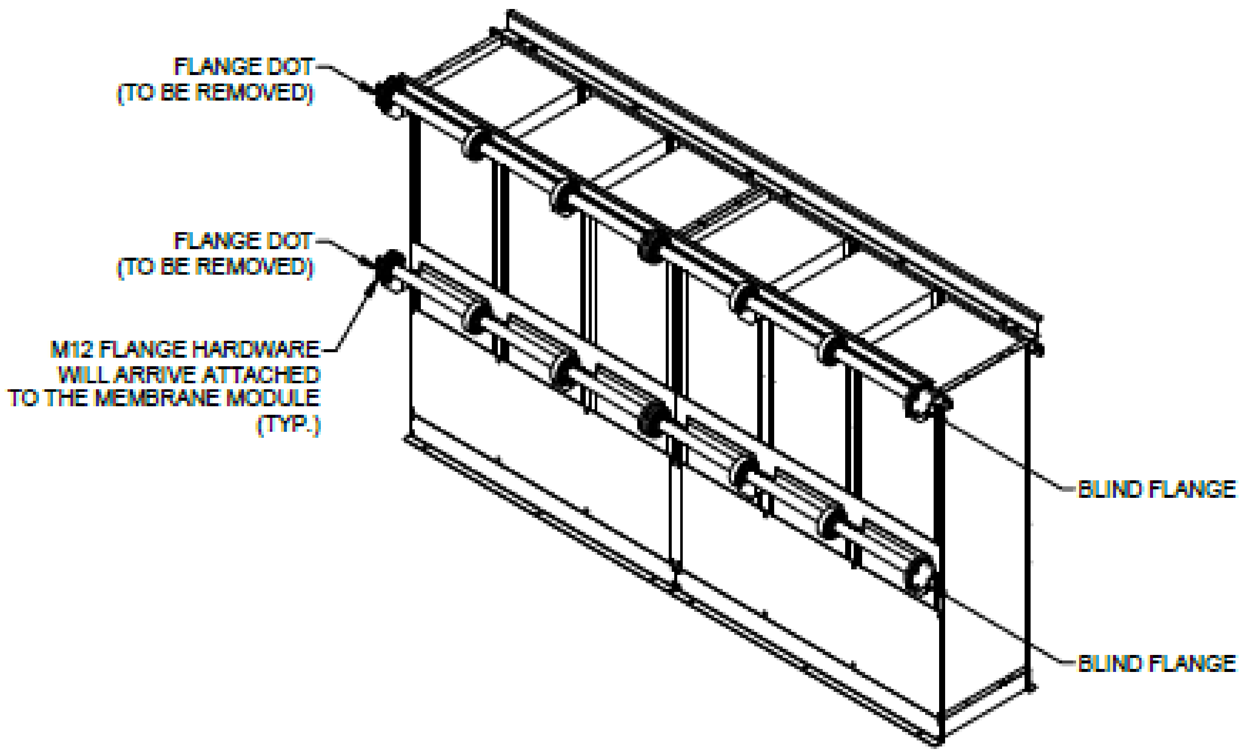
MEMBRANE MODULE INSTALLATION

After

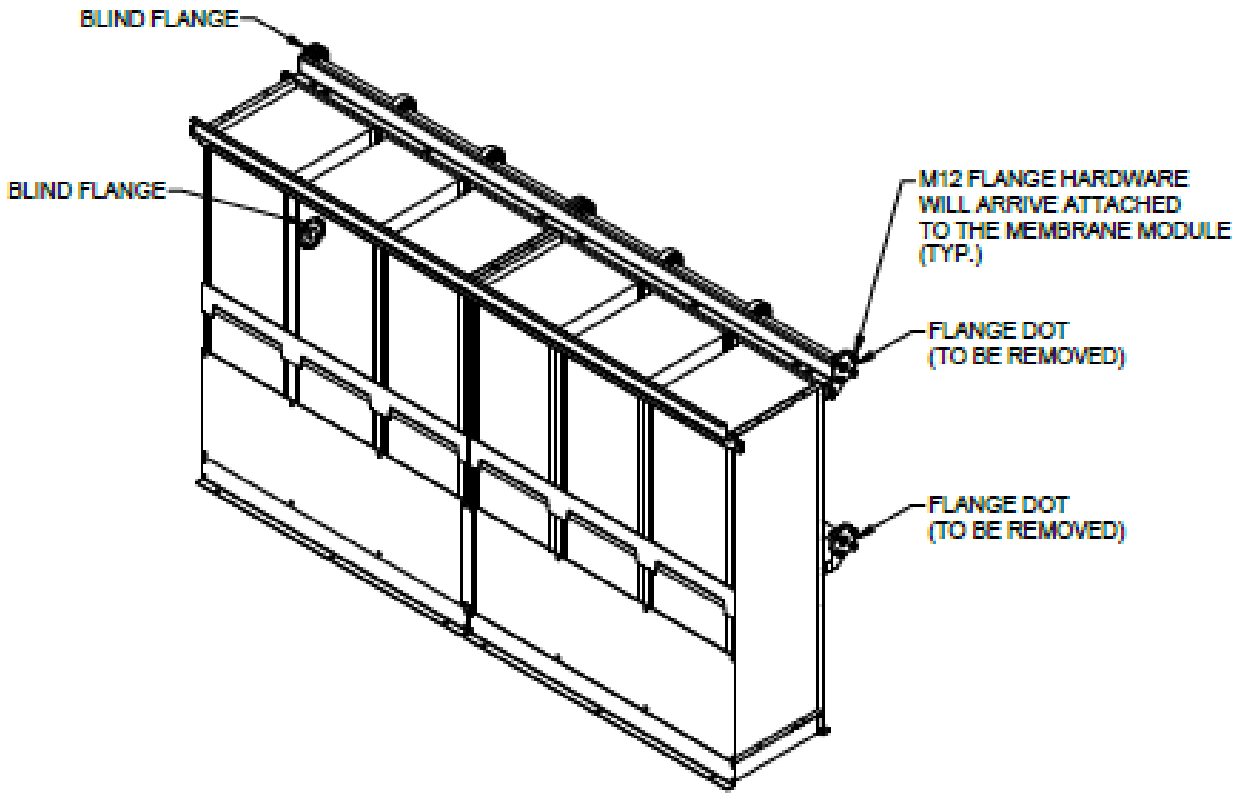
Toray Module

Module Supply – As-Shipped

1 Toray Upper Membrane Module + 1 Toray Lower Membrane Module = 1 Kubota RW400

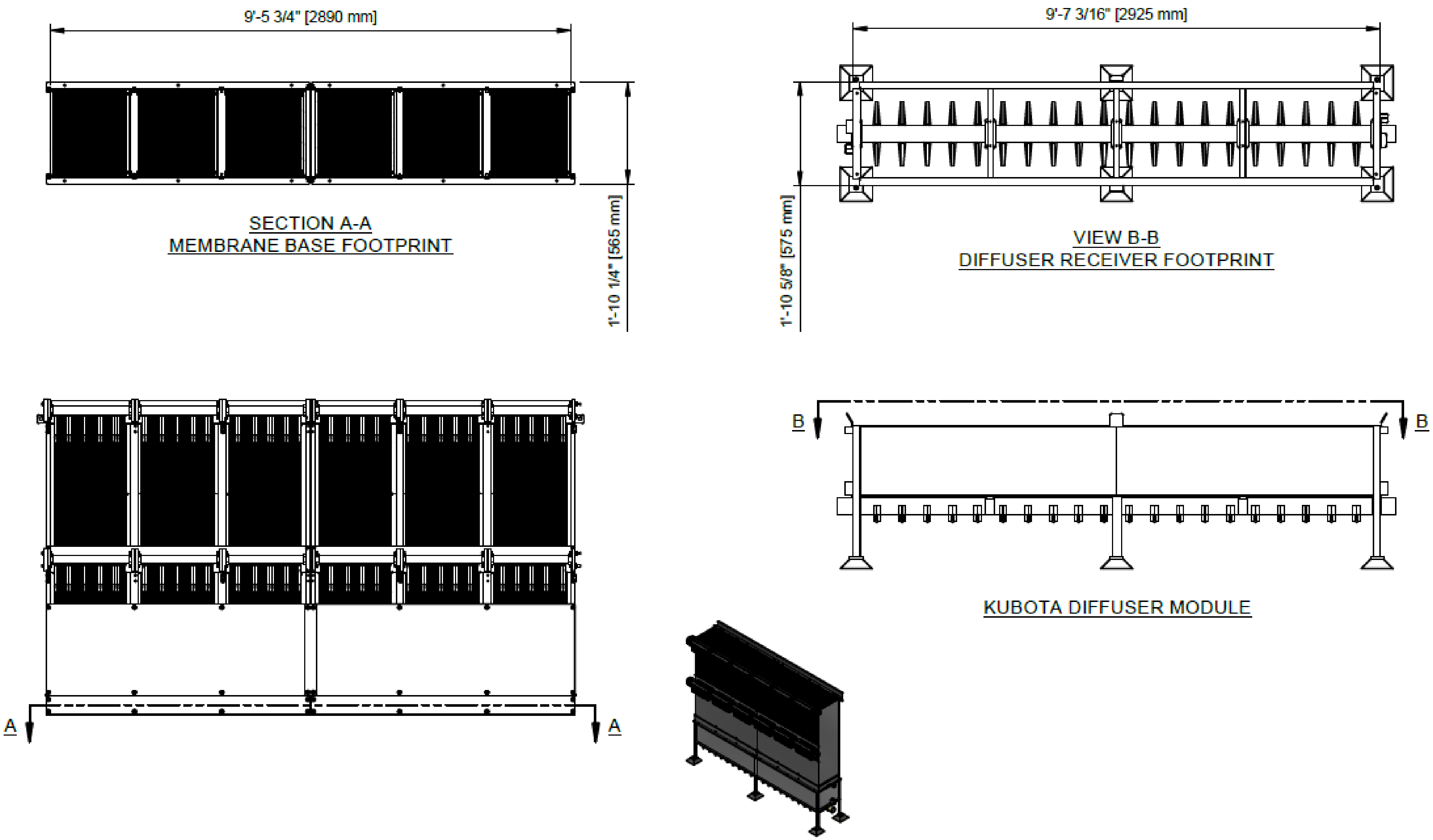


UPPER MEMBRANE MODULE AS-SHIPPED



LOWER MEMBRANE MODULE AS-SHIPPED

Toray ECS210-1200K fit on Kubota Diffuser Module

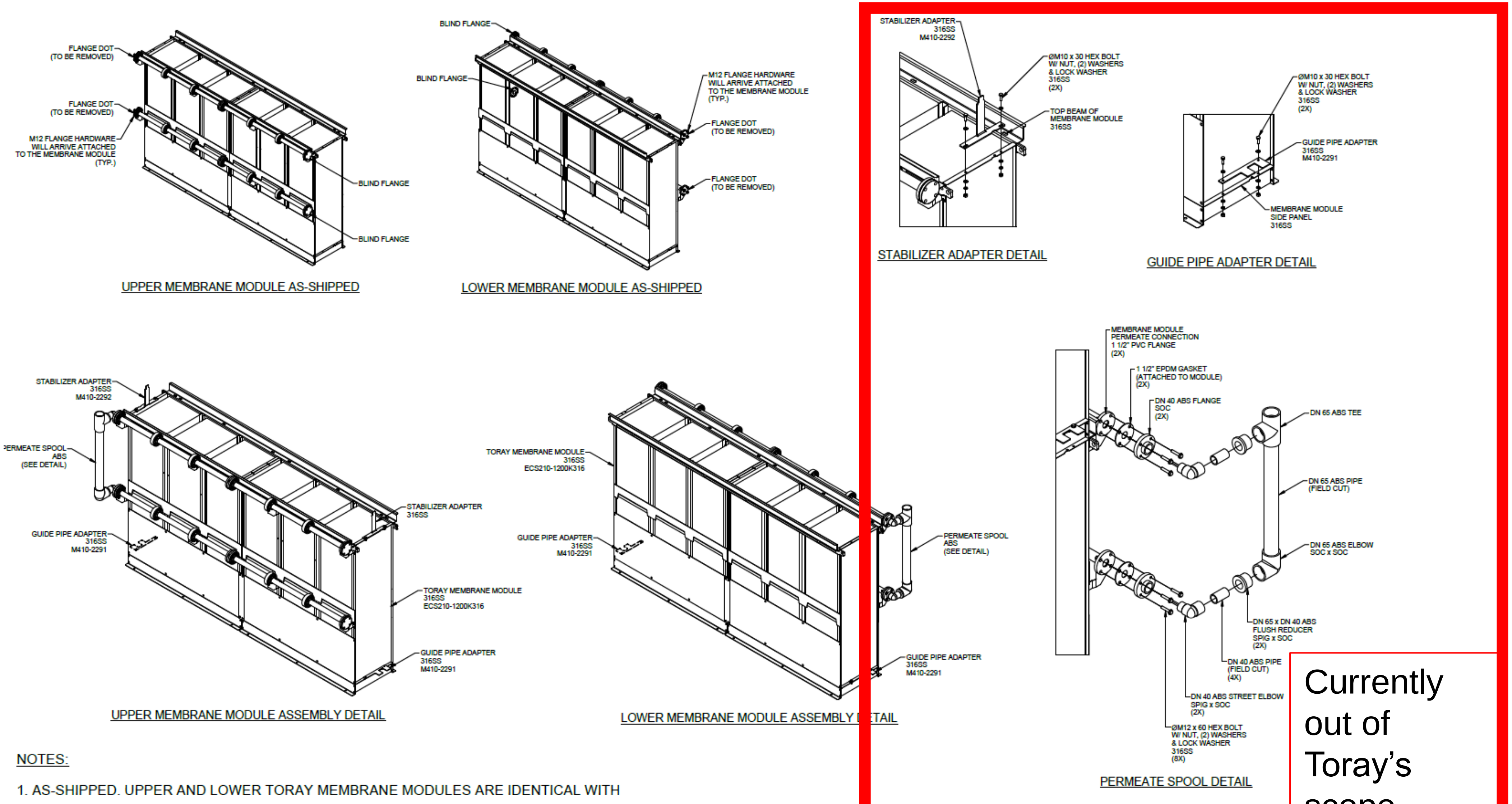


SECTION A-A
MEMBRANE BASE FOOTPRINT

VIEW B-B
DIFFUSER RECEIVER FOOTPRINT

KUBOTA DIFFUSER MODULE

Toray – Scope of Supply



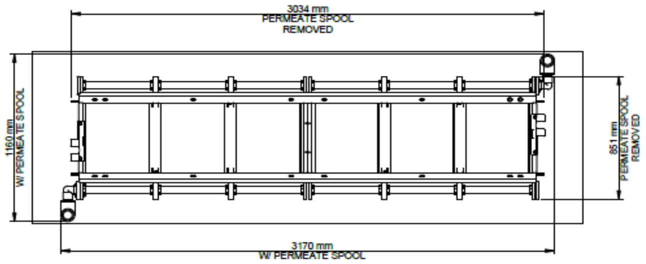
Currently out of Toray's scope.

NOTES:

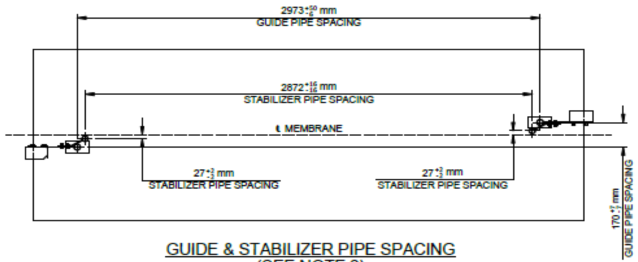
1. AS-SHIPPED. UPPER AND LOWER TORAY MEMBRANE MODULES ARE IDENTICAL WITH

'TORAY'

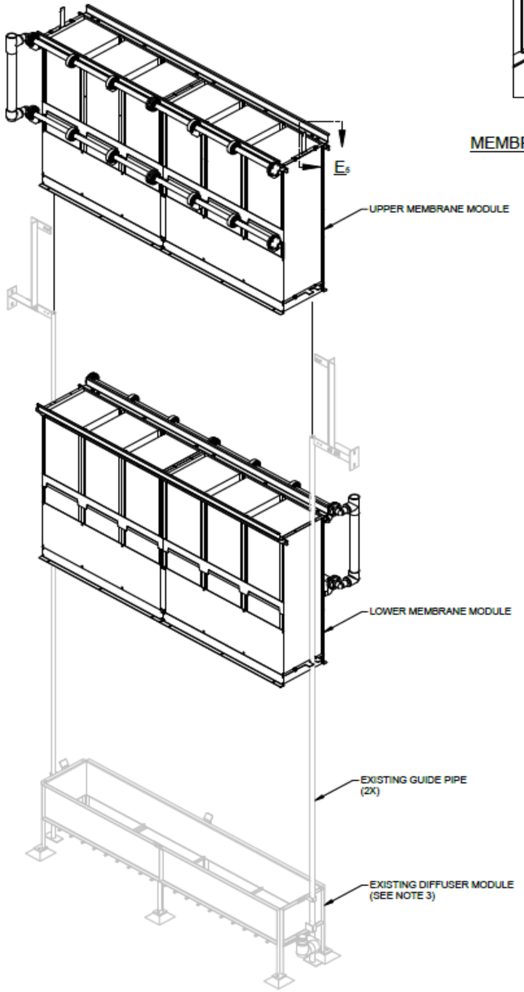
ALWAYS LIFT UPPER & LOWER MEMBRANE MODULES SEPARATELY FROM ONE ANOTHER.



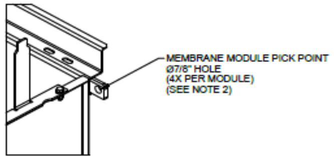
MEMBRANE MODULE LIFTING ENVELOPE



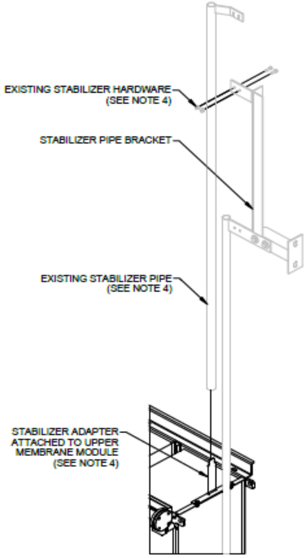
GUIDE & STABILIZER PIPE SPACING
(SEE NOTE 3)



MEMBRANE MODULE INSTALLATION



DETAIL E5
MEMBRANE PICK POINT

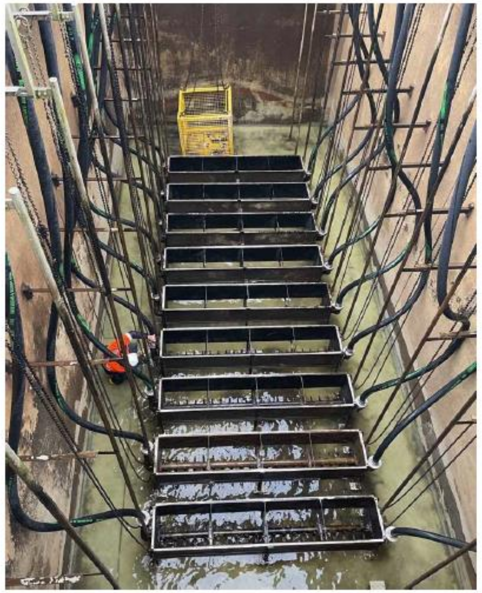


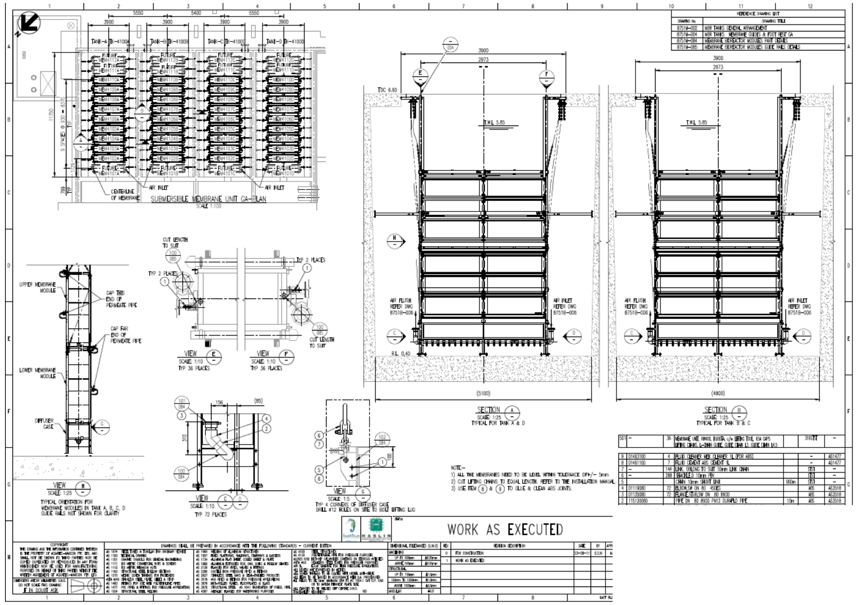
STABILIZER DETAIL
(SEE NOTE 4)

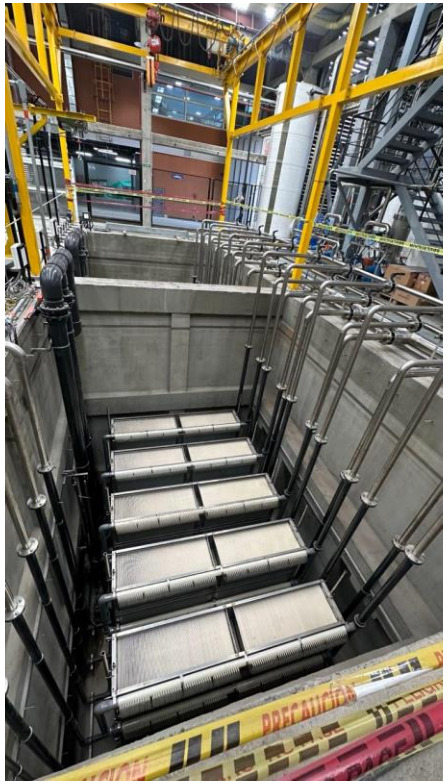
NOTES:

1. ALL NEW PIPING AND EQUIPMENT SHADED DARKER.
2. CAPACITY OF LIFTING FACILITIES MUST BE MORE THAN 'SLUDGED' MASS OF MEMBRANE MODULE (APX. 4400LBS.) PLUS LIFTING TOOL.
3. ENSURE GUIDE PIPE & STABILIZER PIPE SPACING ARE WITHIN TOLERANCES SHOWN.
4. EXISTING STABILIZER PIPES WILL BE REMOVED PRIOR TO REMOVAL OF THE EXISTING MEMBRANE MODULES. ONCE NEW MEMBRANE MODULES ARE INSTALLED, POSITION THE EXISTING STABILIZER PIPES ONTO EACH STABILIZER ADAPTER AND REATTACH TO THE STABILIZER PIPE BRACKET USING THE EXISTING HARDWARE.

Information Request







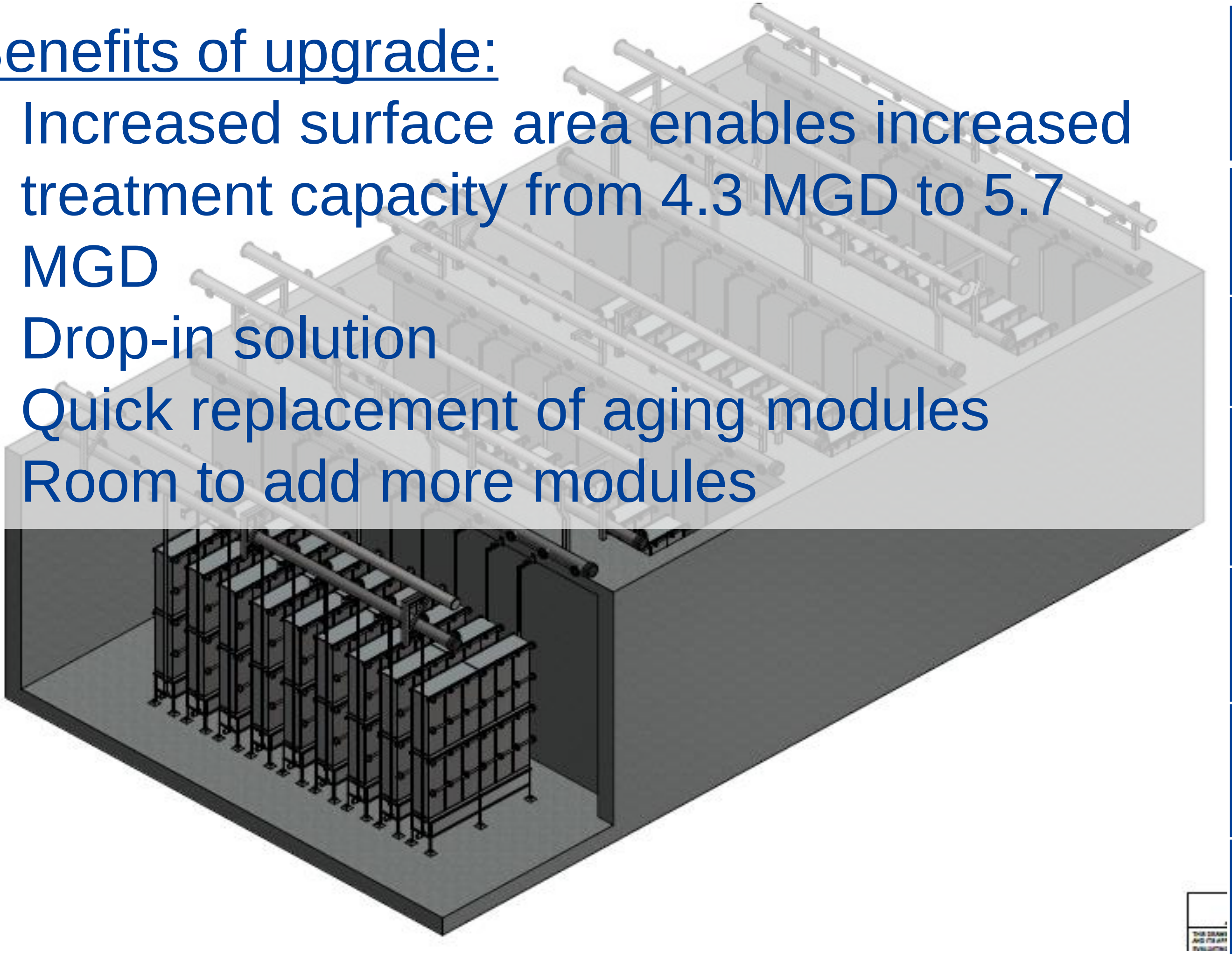
Information Needed:

- Drawings and pictures of membrane tank (examples here)
- Existing filtrate piping sizes
- Lifting tool drawings / pictures

- None!

- Assuming the system has performed well, there is no need to change the operating process.

- If system has not performed well, process changes can be implemented to optimize performance.



Benefits of upgrade:

- Increased surface area enables increased treatment capacity from 4.3 MGD to 5.7 MGD
- Drop-in solution
- Quick replacement of aging modules
- Room to add more modules

		Existing (Competitor K)	Toray Upgrade
MBR Module		RW400	ECS210-1200K
Number of modules	No.	36	36
Total Membrane Surface Area	m^2	20,880	30,240
Airflow per Train	Nm^3/h		Match Existing
Sludge Recirculation Flow	L/s	333-1000	Match Existing
Dimensions of each MBR tank (including spare spaces)	L (m)	11.75	Use existing tanks
	W (m)	4.8	

Kubota RW400 Replacement

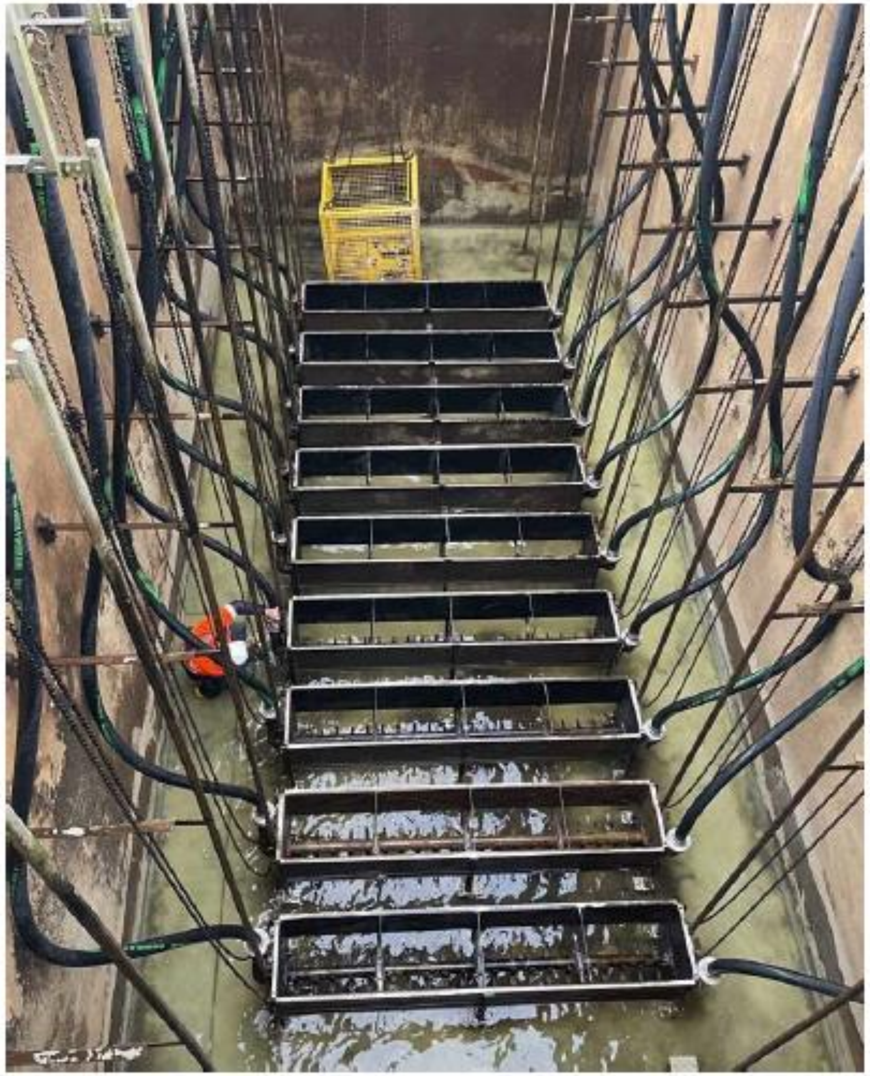
Before

Element blocks removed

1. Prepare Toray modules for installation.
    1. Connect filtrate piping
    2. Install guide plates
    3. Install stabilizer guide pipes

2. Remove Kubota modules

3. Clean out membrane tank



Clean tank with new hoses
(too long)

3. Clean and repair Kubota diffusers

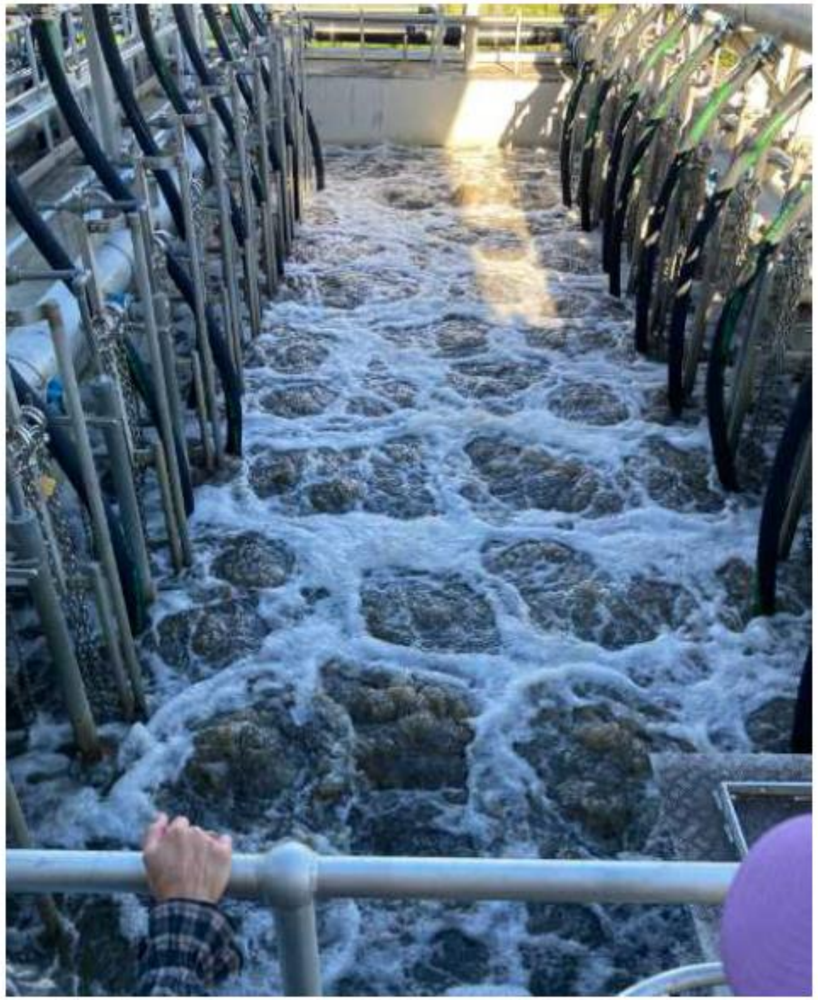
4. Fix leaks in air scour hoses by replacing hoses

5. Check air scour uniformity

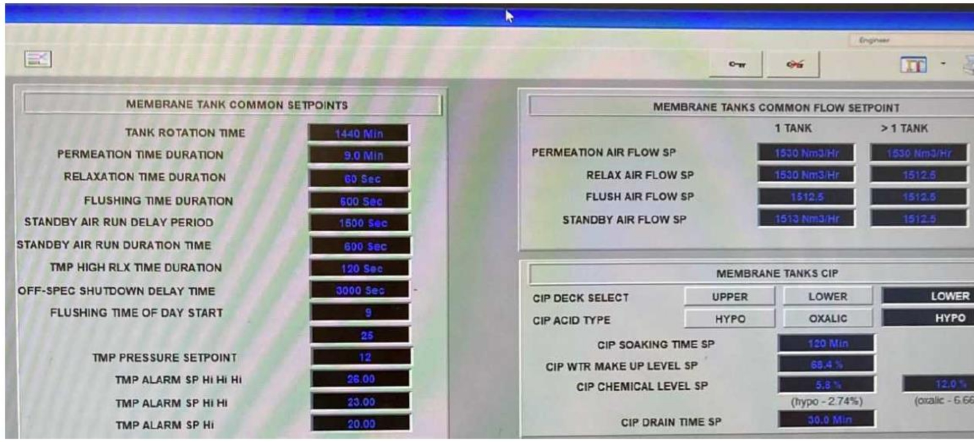
Air scour uniformity testing

6. Place Toray element blocks
7. Reconnect filtrate piping
8. Clean water test
9. Check/revise time settings in SCADA
10. Resume normal operation

Installing Toray element blocks

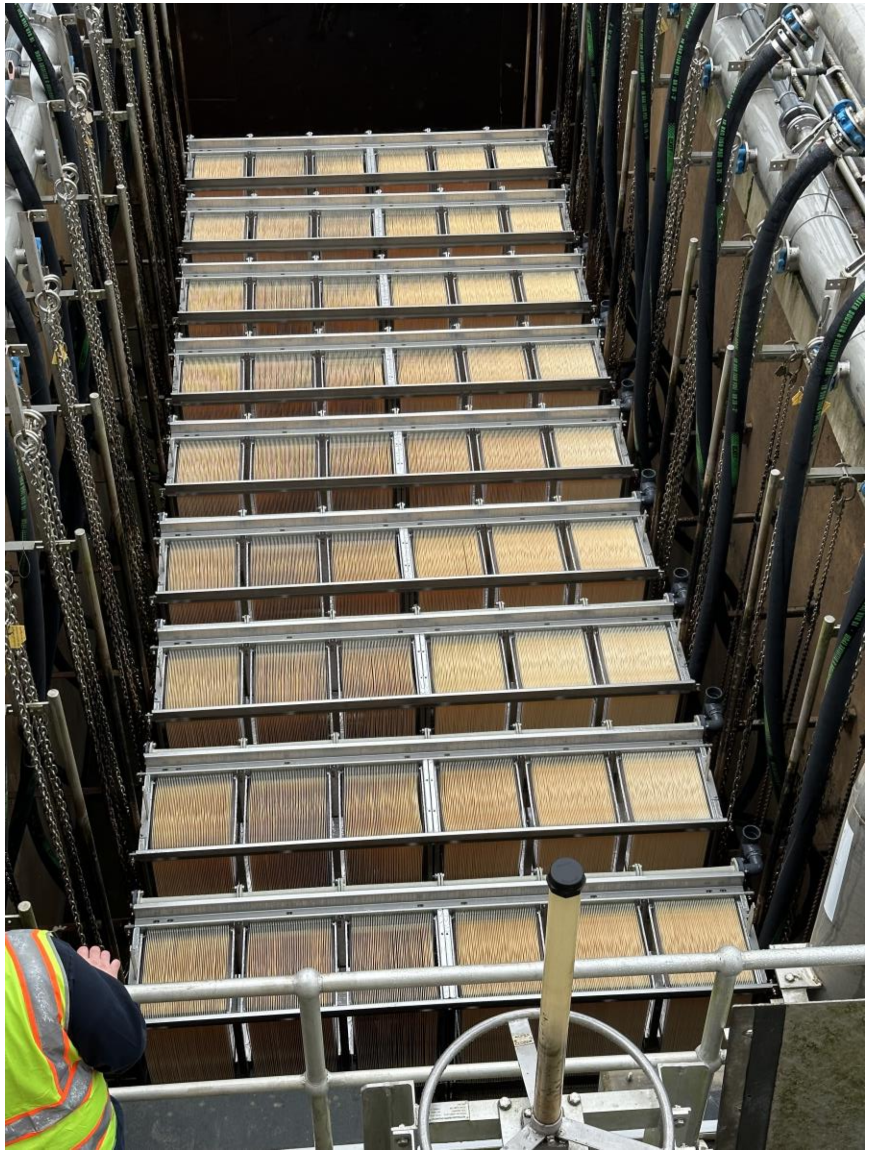


Clean water testing



Revised settings

Kubota RW400 Replacement



After

- ✓ Entire Replacement took ~4 days
- ✓ Kubota module removal, diffuser repair, Toray module installation ~ 1 day.
- ✓ Toray module installation – 9 minutes per module.
- ✓ Initial Performance data within customer's expectations